RUBATO



Because music happens between the lines

User Guide

Give your MIDI a pulse

Version 1.4.0 • Windows 10 / 11 (64-bit)

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1. Meet Rubato

Because music happens between the lines

The digital canvas is too perfect. Every note hits the grid at the exact millisecond and at the exact same volume, and your track loses its heartbeat. Virtual guitars sound like calculators, and synthesized saxophones lose their breath.

Rubato is a Windows utility designed to give your digital instruments a pulse. Rubato (from the Italian *tempo rubato*, meaning “stolen time”) is a musical term for the flexible, expressive shifting of tempo within a phrase. By injecting intelligent micro-timing deviations, natural velocity fluctuations and organic acoustic nuances, Rubato breaks your MIDI out of the machine and infuses it with raw human emotion. **Don’t just play the notes. Let them breathe.**

Why choose Rubato?

- **Organic fluidity:** instantly convert rigid, blocky MIDI data into flowing, expressive musical phrasing.
- **Instrument-specific nuance:** humanization tailored to the physical characteristics of guitars, brass, keys, strings and winds. Rubato ships with a library of instrument presets (section 5).
- **Micro-groove control:** dial in exactly the amount of human imperfection your mix needs, from a tight studio session to a loose, soulful jam.

The anatomy of expression

Rubato goes far beyond simple random shuffling. It recreates the physical, human elements that make a live performance spellbinding.

- **Intelligent micro-timing.** No human drummer or guitarist hits a grid perfectly. Rubato introduces small, intentional rushing and dragging, giving your tracks an authentic pocket and groove.
- **Acoustic velocity curves.** Real musicians accent downbeats and soften transitions. Rubato shapes velocities along the musical phrase and removes the harsh “machine-gun” effect.
- **Instrument-specific articulation.** A saxophone player needs to breathe; a guitarist has natural finger friction against the frets. Rubato mimics these constraints: breath gaps, scoops, glides, falls and strums.
- **The “Human Imperfection” dial.** Complete control over the chaos. Dial it back for a surgically precise result, or push it forward for the raw energy of a live garage band.

How it works in three steps

1. **Import your MIDI.** Open or drag in a quantized MIDI file (or a saved Rubato project).
2. **Choose the voice.** Pick an instrument preset, such as the expressive bend of a blues guitar or the sweeping dynamics of a brass section, or build your own chain of effects.
3. **Breathe life into the track.** Adjust the expression parameters while you listen, then export the new, organic performance back into your DAW.

Rubato does not touch your project tempo. While individual notes flex, rush or drag to create an emotional pocket, the musical phrases stay anchored to the song’s tempo map, so tracks never drift out of time.

Is Rubato free?

Yes. Rubato is 100% free: you can use it forever, with no limitations or payments. It is free software released under the **GNU General Public License, version 3 or later**, and the source code is available on GitHub. If you would like to contribute, you are welcome to fork the project, make your changes and share them with the world.

System requirements

Rubato is a standalone program, not a plug-in. It is lightweight on CPU and RAM and runs smoothly on Windows 10 and Windows 11 (64-bit), leaving all your computing power for your virtual instruments. It may run on earlier versions of Windows, but that has not been tested. It works with any DAW that imports standard MIDI files, including FL Studio, Ableton Live, Cubase, Studio One, Reaper and Pro Tools.

2. Quick start

When you start Rubato, the splash screen appears for a moment (click it to skip) and the main window opens.



The Rubato splash screen. This is the actual artwork used by the program.

Your first rubato in six steps

1. **Open a MIDI file.** Choose *File* > *Open MIDI file or project* (Ctrl+O), or drag a `.mid` file onto the window. A format 0 file (all channels in one track) is split into one track per channel automatically.
2. **Choose where the sound comes from.** In the *MIDI out* box (top left) pick the synthesizer or virtual MIDI port that should play the song.
3. **Select a track** by clicking its row, then press its **FX** button (or press Ctrl+E, or double-click the track strip). The **MFX Chain** window opens.
4. **Load a preset or build a chain.** Click *Presets* and choose an instrument, for example “Flute - legato breathy”, or press *Insert* to add effects one by one.
5. **Listen and adjust.** Press the green Play button (or the Space bar). The graphs change live and playback restarts automatically whenever you change a parameter. Use *Bypass FX* for an A/B comparison with the original.
6. **Keep the result.** Press **Apply All FXs** to write the effects into the song, then *File* > *Export MIDI file* (Ctrl+Shift+S). Save the project (Ctrl+S) to keep your effect chains for later.

> **Nothing is permanent until you say so.** The effects are a live preview. The song itself is not changed until you press **Apply All FXs**, and **Undo** (Ctrl+Z) restores the song to the state before the last Apply.

3. The main window

The main window shows every track of your song as a coloured strip with its notes, with the controller graphs drawn on top, one strip per track. The figure below shows a real song (a project with five effects on track 2); the numbers are explained underneath.

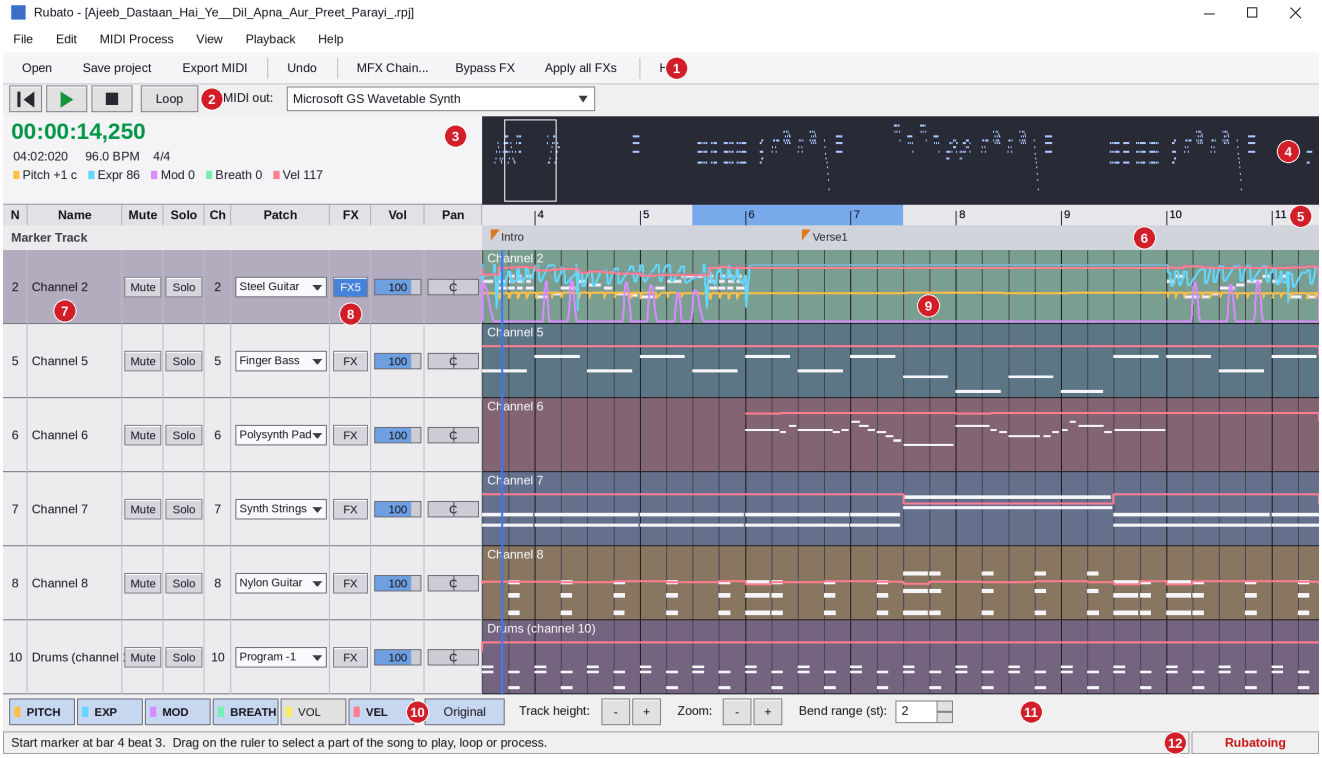


Figure 1 - The main window (illustration rendered from the program's layout using real project data). Numbers refer to the list below.

Parts of the window

#	Part	What it does
1	Toolbar	Open, Save project, Export MIDI, Undo, MFX Chain..., Bypass FX (toggle), Apply all FXs and Help. The same commands are in the menus.
2	Transport	Go to beginning, Play (from the start marker, or only the selected part), Stop and Loop. The *MIDI out* box chooses the device that plays the song. The Space bar also plays and stops.
3	Information panel	Shows the position as time and as bar:beat:tick, the tempo and time signature, and the value of each visible graph of the selected track at the playhead (or under the mouse). Each value has a colour chip matching its graph.
4	Overview	A miniature of the whole song. The white frame is the part you see below; click anywhere in the overview to jump there.
5	Ruler	Bar numbers. Click to set the start position; drag to select a part of the song (shown in blue). Play then plays only the selection.
6	Marker Track	Text markers. Double-click to add one, double-click a marker to edit it, right-click for the menu, Delete removes the selected marker.
7	Track list	One row per track: number, name, Mute, Solo, MIDI channel (Ch), Patch (instrument), FX, Vol and Pan. Selected track is highlighted.
8	FX button	Opens the MFX Chain window of the track. It shows the number of effects in the chain (FX5 = five effects) and turns blue when the chain is active.
9	Track strips	Notes (white bars) and controller graphs. On the selected track, the original is drawn as grey "ghost" notes and grey lines behind the processed result.

#	Part	What it does
10	Graph buttons	PITCH, EXP, MOD, BREATH, VOL and VEL show or hide each graph (see the colour table below).
11	Original / size / zoom / bend	*Original* toggles the ghost display; *Track height* and *Zoom* have - and + buttons; *Bend range* is the pitch-bend range of your synthesizer in semitones.
12	Status line and indicator	The status line explains the last action. The box at the right reads **Original** when no track has an effect in progress and a red **Rubatoing** when the preview differs from the original.

The graph colours

Button	Graph	MIDI controller	Colour
PITCH	Pitch bend	Pitch wheel	orange
EXP	Expression	CC 11	light blue
MOD	Modulation	CC 1 (mod wheel)	violet
BREATH	Breath	CC 2	green
VOL	Channel volume	CC 7	yellow
VEL	Note velocity	(per note)	pink

The result lines in the MFX Chain window (section 5) use these same colours, so a number is always easy to match with its graph.

Working with tracks and the timeline

Track list

- **Mute / Solo** work while the song is playing.
- **Vol and Pan:** drag the sliders; double-click a slider to reset it.
- **Patch:** click the Patch box to choose another General MIDI instrument for the track. A changed patch is shown in orange; *As in the file* undoes the change.
- **Ch:** click the Ch box to move the track to another MIDI channel.
- **Column widths:** drag the dividers in the header row.
- **Delete a track:** right-click it. Rubato asks first and *Undo* brings the track back.

Selecting a part of the song

Drag across the ruler to select a few bars. **Play** then plays only that part, and **Loop** repeats it, so you can judge an effect on a short passage. Switch Loop off while it plays and the music simply continues to the end. Click the ruler to clear the selection. In the MFX Chain window, *From selection* turns the selection into the From/To bars of an effect.

Markers

The Marker Track is the thin row at the top of the strips. Markers hold text (letters and digits only) and are written into the exported MIDI file, so they appear in your DAW. Add one with a double-click, or with *Edit* > *Insert marker at the start position*.

Mouse and keyboard

Action	How
Scroll the tracks	Mouse wheel
Scroll in time	Shift + wheel, or a sideways swipe on a touch pad
Zoom in time	Ctrl + wheel, or the Zoom - / + buttons
Set the start position	Click the ruler, or click a track strip (blue line)
Play / stop	Space bar (clicking the ruler while playing continues from there)
Open a file	Ctrl+O, or drag a file onto the window
Save project	Ctrl+S
Export MIDI file	Ctrl+Shift+S
Undo	Ctrl+Z
Open the MFX Chain	Ctrl+E, the FX button, or double-click the track strip
Quick guide	Help menu

4. Menus at a glance

Menu	Commands
File	Open MIDI file or project (Ctrl+O) - Open project (*.rpj) - Open recent - Save project (Ctrl+S) - Save project as - Export MIDI file, format 0 (Ctrl+Shift+S) - Exit
Edit	Undo (Ctrl+Z) - Insert marker at the start position
MIDI Process	MFX chain of selected track (Ctrl+E) - Copy chain of selected track to all tracks - Bypass all FX (A/B) - Apply all FXs to the song
View	Show or hide each graph - Show original (ghost) while editing - Taller tracks / Shorter tracks - Zoom in / Zoom out
Playback	Play from start marker - Stop - Go to beginning - Clear timeline selection
Help	Quick guide (a resizable window with scroll bars) - About (version, licence, links)

Files Rubato uses

Extension	What it is
.mid / .midi	Standard MIDI files. Rubato reads them and always exports **format 0** (a single track), exactly as you see and hear it.
.rpj	Rubato project: the song, the effect chains of every track, the mixer settings (volume, pan, patch, mute, solo, channel), the markers and the view settings.
.rps	Rubato preset: one effect chain, saved from the MFX Chain window. The older extension .mfxp is still read.

5. The MFX chain window

Every track has its own **chain** of effects. The MFX Chain window is where you build it. It stays open next to the main window, so you can watch the graphs while you change a parameter.

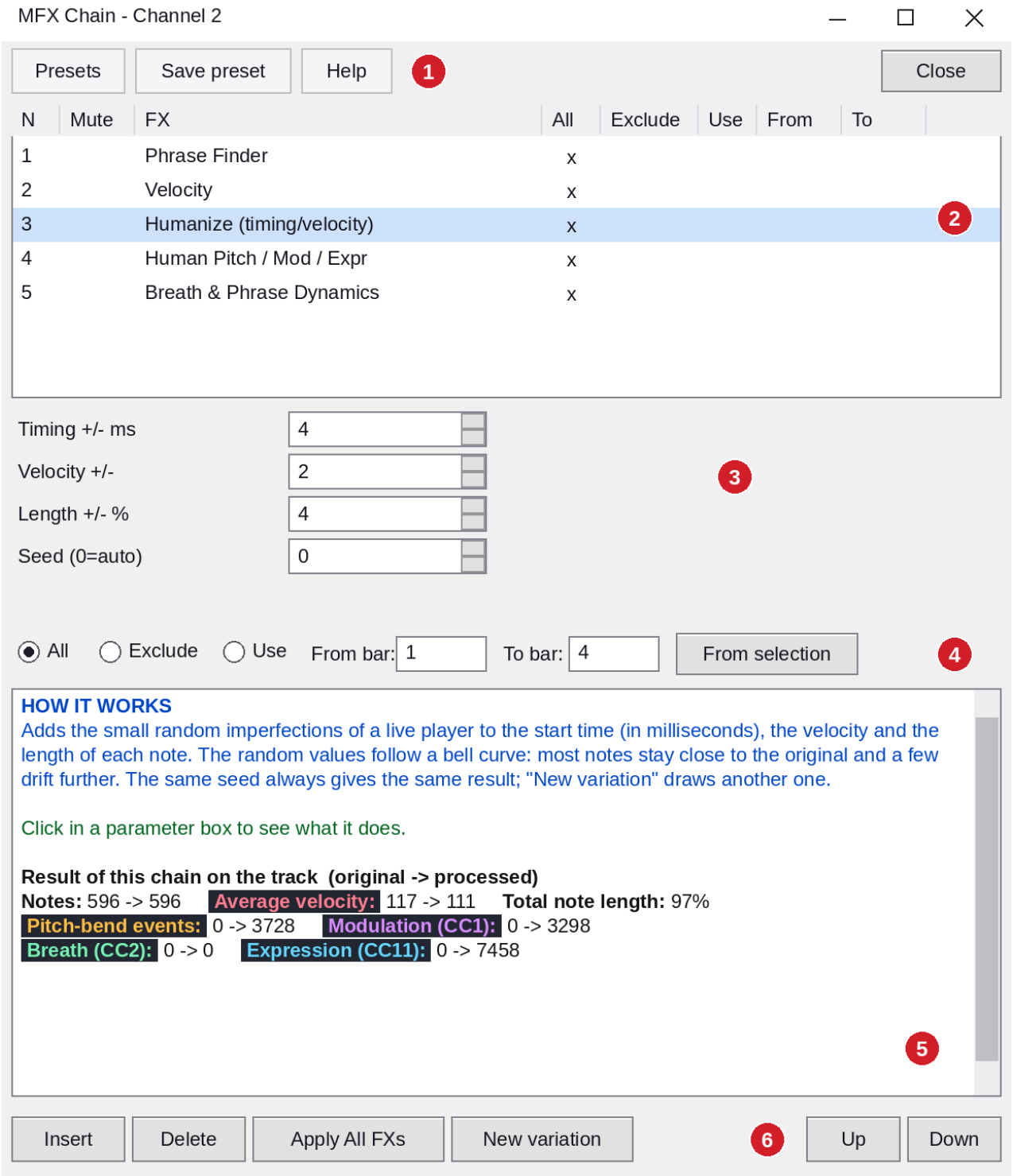


Figure 2 - The MFX Chain window with a five-effect chain. Numbers refer to the list on the next page.

Parts of the MFX chain window

#	Part	What it does
1	Toolbar	**Presets** opens the preset menu (factory instruments grouped by family, your own presets, and *Open preset file*). **Save preset** stores the current chain as a *.rps file. **Help** opens the help text.
2	Effect list	The chain, in order, from the top. Columns: **N** (position), **Mute** (an x switches the effect off without deleting it - click the cell to toggle), **FX** (the effect), and the range columns **All / Exclude / Use** with **From** and **To** bars.
3	Parameters	The settings of the selected effect: up to two columns of five boxes. Type a value or use the spin buttons (Shift + click moves in bigger steps). Values outside the allowed range are limited automatically.
4	Range	Choose where the effect applies: **All** = the whole song; **Use** = only inside the From/To bars; **Exclude** = everywhere except the From/To bars. **From selection** copies the bars of your ruler selection.
5	Description box	Explains how the selected effect works (blue) and, when you click in a parameter box, what that parameter does (dark green). Below it, a summary of the whole chain on the track, with each name in the colour of its graph.
6	Buttons	**Insert** adds an effect after the selected one; **Delete** removes it; **Apply All FXs** writes all chains into the song; **New variation** draws another random result for the selected effect; **Up / Down** move the effect in the chain.

Reading the result summary

The bottom of the description box tells you what the chain did to the track, as *original -> processed*: the number of notes, the average velocity, the total note length as a percentage of the original, and the number of pitch-bend, modulation (CC1), breath (CC2) and expression (CC11) events. If you see, for example, *Pitch-bend events: 0 -> 3728*, the chain has drawn about three and a half thousand pitch-wheel movements that the original did not have.

> **Copy a chain to all tracks.** After setting up one track, use *MIDI Process > Copy chain of selected track to all tracks*, then adjust the individual tracks. Phrase and breath effects make most sense on single-line instruments (winds, strings, vocals, lead guitar); chords and drums usually need only timing and velocity effects.

Presets

Rubato comes with a library of factory presets, grouped by instrument family. Several are marked *(matches the reference sample)*: they were tuned against recordings of real players.

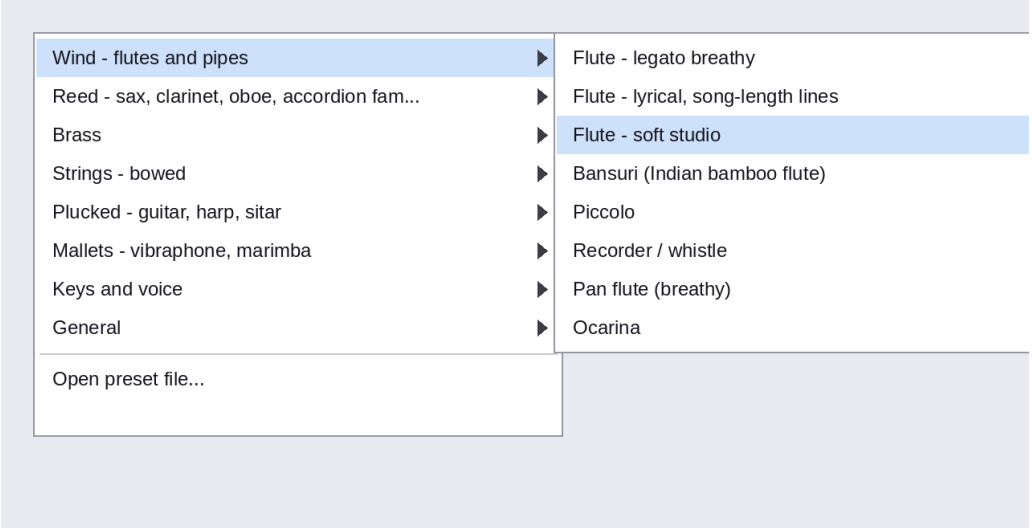


Figure 3 - The Presets menu (illustration). Your own presets appear under User presets.

Family	Examples
Wind - flutes and pipes	Flute (legato breathy, lyrical, soft studio), Bansuri, Piccolo, Recorder / whistle, Pan flute, Ocarina
Reed	Alto, tenor, soprano and baritone sax, Clarinet, Oboe, English horn / bassoon, Harmonica, Shehnai, Accordion / harmonium
Brass	Trumpet, Trombone, French horn, Brass section
Strings - bowed	Violin, Viola, Cello, Contrabass, String ensemble, Pizzicato
Plucked	Guitar solo and rhythm styles, Nylon / acoustic guitar, Bass guitar, Sitar (meend slides), Harp
Mallets	Vibraphone
Keys and voice	Piano, Organ, Choir / voice oohs, Vocal - Indian gamaka style
General	Any instrument (gentle, or loose and laid-back), Tidy up: quantize to 1/16 (60%), Swing the eighth notes

To keep your own sound, build a chain, press *Save preset* and give it a name. Presets are stored as `.rps` files in the `Presets` folder next to the program and are listed under *User presets*. You can share the files with friends.

6. Order matters: how a chain is evaluated

A chain is evaluated **from the top to the bottom**. Each effect receives the notes and controllers that the effect above it produced, and passes its own result on to the next. That is why the same effects in a different order can sound different.

Example: velocity

Take a note with velocity 100. The **Velocity** effect calculates *new* = *old x Scale%* + *Offset*, limited to 1-127. **Humanize** adds a random amount, limited to the *Velocity +/-* setting (a bell curve, so most notes move less than the maximum), again limited to 1-127.

Chain order	Step 1	Step 2	Result
Velocity 80%, then Humanize +/-10	100 x 0.8 = 80	80 +/- up to 10	70 to 90, centred on 80
Humanize +/-10, then Velocity 80%	100 +/- up to 10 = 90 to 110	x 0.8	72 to 88, centred on 80
Velocity 80%, then Velocity +10	100 x 0.8 = 80	80 + 10	90
Velocity +10, then Velocity 80%	100 + 10 = 110	110 x 0.8	88

Near the limits the order matters even more. A loud note at 120 followed by Humanize +/-10 can reach 127 and is cut off; if Humanize comes first and the scale comes last, the top of the random range is lost before it can be scaled. Once a value is cut off at 1 or 127, a later effect cannot restore it.

Rules of thumb

- Put the **Phrase Finder first**. It only influences the effects below it (Human Pitch / Mod / Expr and Breath & Phrase Dynamics).
- Put **Transpose** and **Quantizer** early; they tidy the raw material.
- Set the level with **Velocity** before you **Humanize**, so that the random changes are not cut off at the limits.
- Put **Strum** after timing effects, so that the spread of the chord is not undone by them.
- Controller effects (Human Pitch / Mod / Expr, Breath) write pitch bend, modulation, breath and expression. When two effects write the same controller, the one lower in the chain wins for that controller.
- If a result surprises you, switch single effects off with the Mute column to find the one responsible.

> **A note about timing effects.** Controller curves are drawn from the notes as they are at that point of the chain. If you place a timing effect (Quantizer, Humanize, Strum) *below* a controller effect, the notes move but the curves already drawn stay where they were. For the best result, place timing effects above Human Pitch / Mod / Expr and Breath.

7. The effects

Rubato has eight effects. Each is described below with its parameters, default values and allowed ranges. The figures show each effect’s parameter area as it appears with the default settings.

7.1 Transpose

Moves every note up or down by whole semitones (12 = one octave). Notes pushed outside the MIDI range are removed. Drums (channel 10) can be left alone.

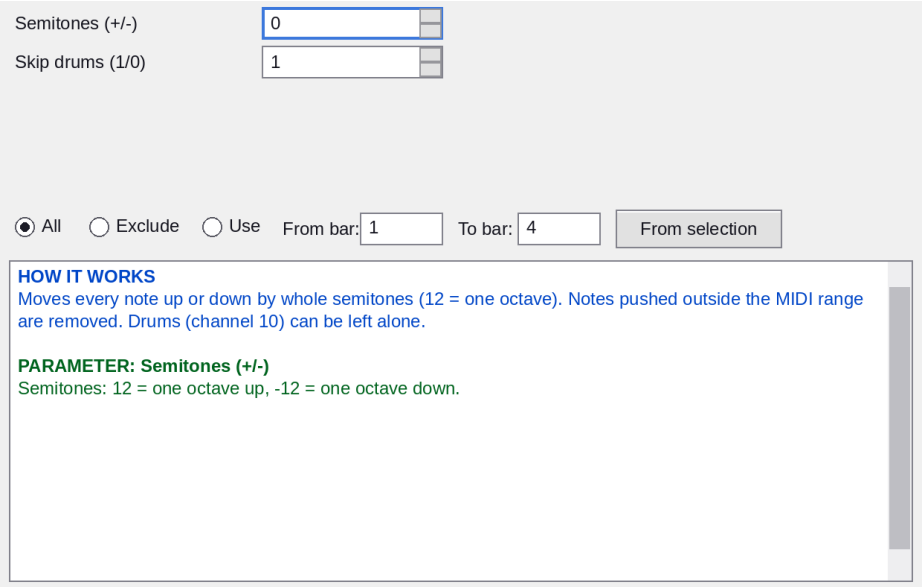


Figure - Transpose, default settings.

Parameter	Default	Range	What it does
Semitones (+/-)	0	-48 to 48	Semitones: 12 = one octave up, -12 = one octave down.
Skip drums (1/0)	1	0 to 1	1 = leave the drum channel (10) untouched, 0 = transpose it as well.

Use it to move a part into the range of the instrument you want, or to try a part in another key.

- 12 = one octave up, -12 = one octave down. Notes pushed outside the MIDI range are removed.
- Drums (channel 10) are left alone by default; set *Skip drums* to 0 to transpose them too.

7.2 Quantizer

Snaps note starts to a rhythmic grid. Strength 100% puts notes exactly on the grid; lower values move them only part of the way and keep some of the original feel. Start offset moves the whole grid, Shift onward delays every second grid point (swing), and the duration settings tidy the note lengths.

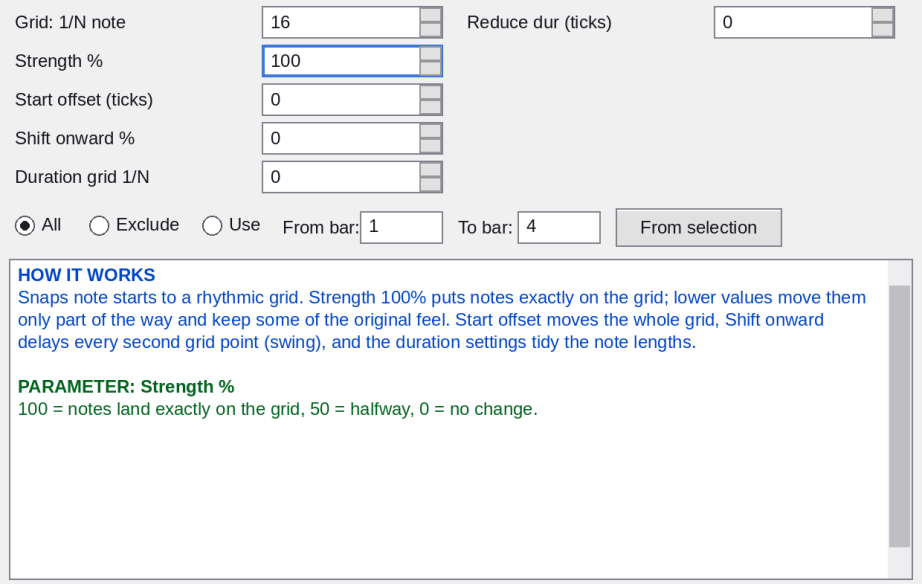


Figure - Quantizer, default settings.

Parameter	Default	Range	What it does
Grid: 1/N note	16	1 to 128	Grid size: 16 = sixteenth notes, 8 = eighth notes, 4 = quarter notes, 32 = thirty-second notes.
Strength %	100	0 to 100	100 = notes land exactly on the grid, 50 = halfway, 0 = no change.
Start offset (ticks)	0	-480 to 480	Moves the whole grid by this many MIDI ticks (positive = later). Small values (2-6) give a laid-back feel.
Shift onward %	0	0 to 100	Delays every second grid point by this % of the grid size (swing / shuffle). 0 = straight.
Duration grid 1/N	0	0 to 128	Rounds note lengths to this grid (16 = sixteenth). 0 = leave the lengths alone.
Reduce dur (ticks)	0	0 to 480	Shortens every note by this many ticks. A value of 1 removes overlaps between neighbouring notes.

Quantizing is the opposite of humanizing, so it is usually used first: either to tidy up a sloppy recording before the human effects, or to add swing.

- **Tidy:** Grid 16, Strength 60% pulls notes most of the way to the sixteenth-note grid while keeping a little of the original feel.
- **Swing:** Grid 8 with *Shift onward* around 33% gives a triplet-like shuffle.
- **Laid-back feel:** a small *Start offset* (2-6 ticks) shifts the whole grid later.

7.3 Humanize (timing / velocity)

Adds the small random imperfections of a live player to the start time (in milliseconds), the velocity and the length of each note. The random values follow a bell curve: most notes stay close to the original and a few drift further. The same seed always gives the same result; "New variation" draws another one.

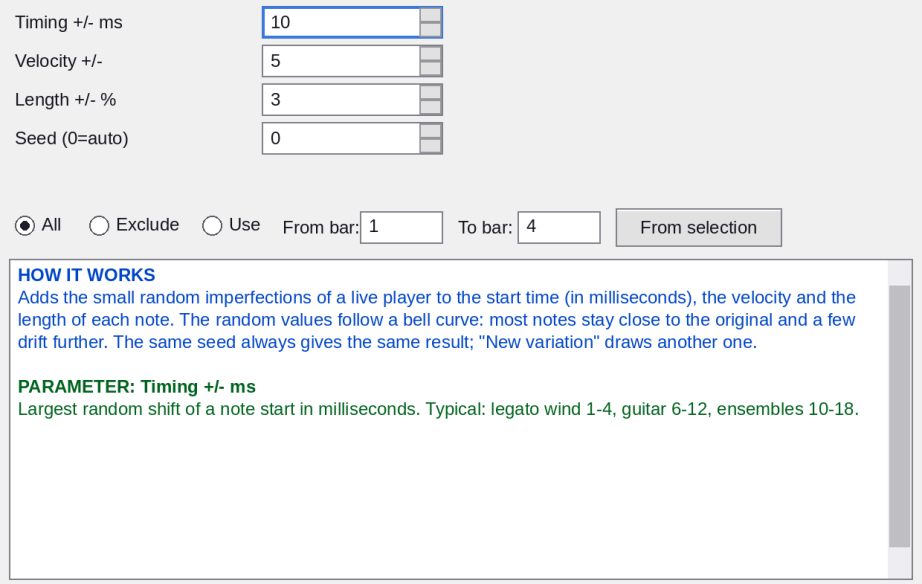


Figure - Humanize (timing/velocity), default settings.

Parameter	Default	Range	What it does
Timing +/- ms	10	0 to 250	Largest random shift of a note start in milliseconds. Typical: legato wind 1-4, guitar 6-12, ensembles 10-18.
Velocity +/-	5	0 to 127	Largest random change of the velocity (loudness) of a note.
Length +/- %	3	0 to 100	Largest random change of the length of a note, in percent.
Seed (0=auto)	0	0 to 999999	0 = automatic (stable for this effect). Any other number gives its own repeatable variation.

This is the basic effect of Rubato: the small random imperfections of a live player.

- Typical timing values (ms): legato winds 1-4, guitar 6-12, ensembles 10-18.
- The same *Seed* always gives the same result. With seed 0 the result is stable for that effect; press **New variation** to draw another one, or type your own number to bring a favourite back.

7.4 Velocity

Scales and then shifts all note velocities: new velocity = old x Scale% + Offset. Use it to bring a flat or too quiet/loud recording to the level a preset expects before humanizing.

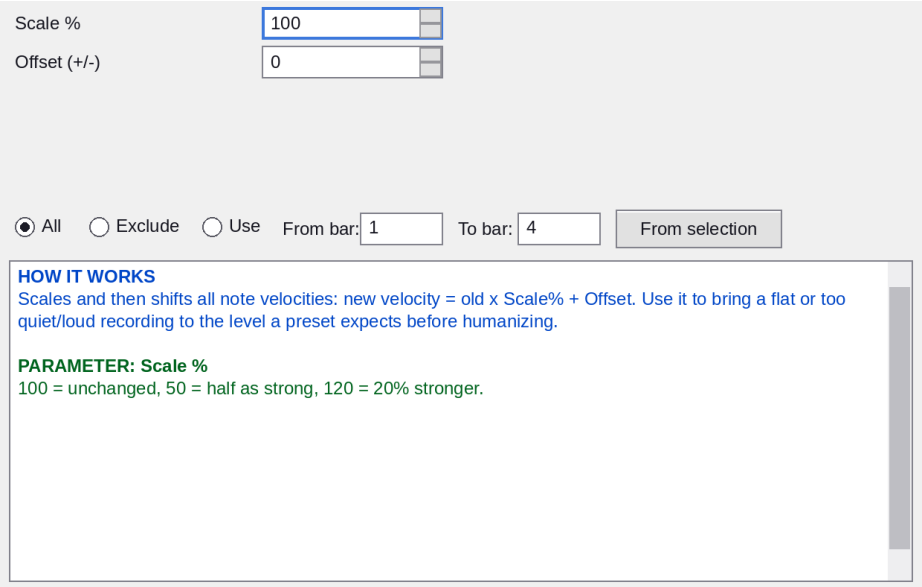


Figure - Velocity, default settings.

Parameter	Default	Range	What it does
Scale %	100	0 to 400	100 = unchanged, 50 = half as strong, 120 = 20% stronger.
Offset (+/-)	0	-127 to 127	Added after scaling. Raw recordings at velocity 82 become 96 with an offset of +14.

A simple level control. Use it to bring a flat, too quiet or too loud recording to the level a preset expects *before* humanizing.

- Scale 50% = half as strong; 120% = 20% stronger. A recording at velocity 82 becomes 96 with an Offset of +14.

7.5 Phrase Finder

Finds the musical phrases - the places where a player would take a breath or lift the bow - and hands them to the effects BELOW it in the chain (Human Pitch/Mod/Expr, Breath & Phrase Dynamics). A phrase is between Min and Max long, measured in sixteenth notes. It ends at a rest of at least 'Pause', otherwise at the best break inside the allowed length, preferring long notes, big pitch jumps and velocity drops. The phrase starts are shown as small red marks on the track strip. Put it first in the chain.

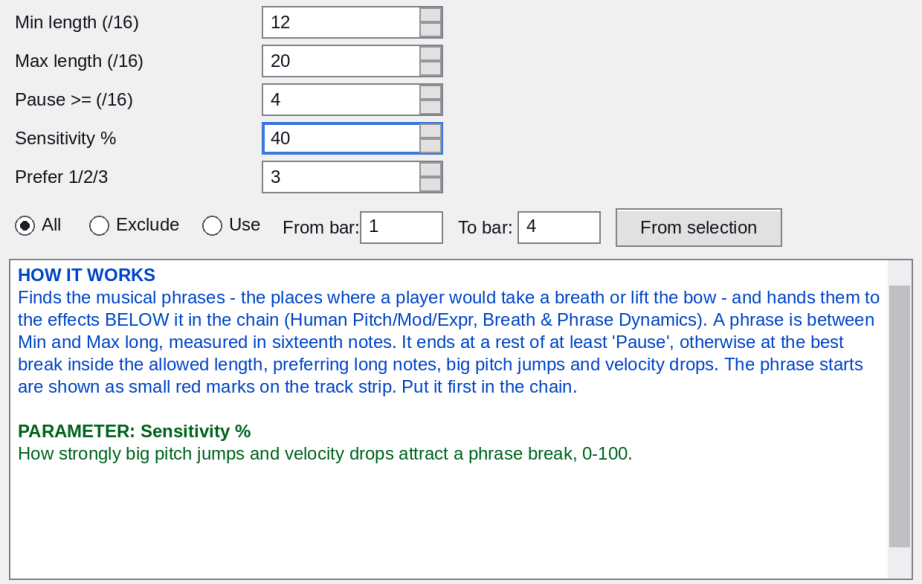


Figure - Phrase Finder, default settings.

Parameter	Default	Range	What it does
Min length (/16)	12	1 to 128	Shortest phrase in sixteenth notes (12 = three beats).
Max length (/16)	20	1 to 256	Longest phrase in sixteenth notes (20 = five beats). The phrase is cut at the best place before this length.
Pause >= (/16)	4	0.5 to 64	A rest of at least this many sixteenth notes always ends a phrase.
Sensitivity %	40	0 to 100	How strongly big pitch jumps and velocity drops attract a phrase break, 0-100.
Prefer 1/2/3	3	1 to 3	What to look at when choosing a break: 1 = velocity drops, 2 = pitch jumps, 3 = both.

The Phrase Finder does not change a single note itself. It finds the musical phrases and gives them to the effects below it.

- The start of every phrase is shown as a small red mark on the track strip, so you can check that Rubato breathes where a player would.
- A phrase is between *Min* and *Max* long, measured in sixteenth notes (12 = three beats, 20 = five beats). A rest of at least *Pause* always ends a phrase.
- Raise *Sensitivity* if you want pitch jumps and velocity drops to attract phrase breaks more strongly; *Prefer* chooses what to look at (1 = velocity drops, 2 = pitch jumps, 3 = both).
- The Phrase Finder always looks at the whole song; its range setting is ignored.

7.6 Human Pitch / Mod / Expr

Draws the pitch-wheel and modulation movement of a live player. Drift: slow tuning wander. Vibrato: pitch vibrato that grows on long notes. Scoop-in: a note starts slightly flat and slides up into pitch - 'Scoop on' chooses which notes (wind players scoop long notes approached from below, guitarists get a pick dip on every note). Glide: finger/fret slide into upward leaps. Fall-off: the pitch drops at the end of a phrase. Modulation: a vibrato-depth bell on long notes, for sounds that use the mod wheel. The pitch-bend range of your synth is set in the box at the bottom right.

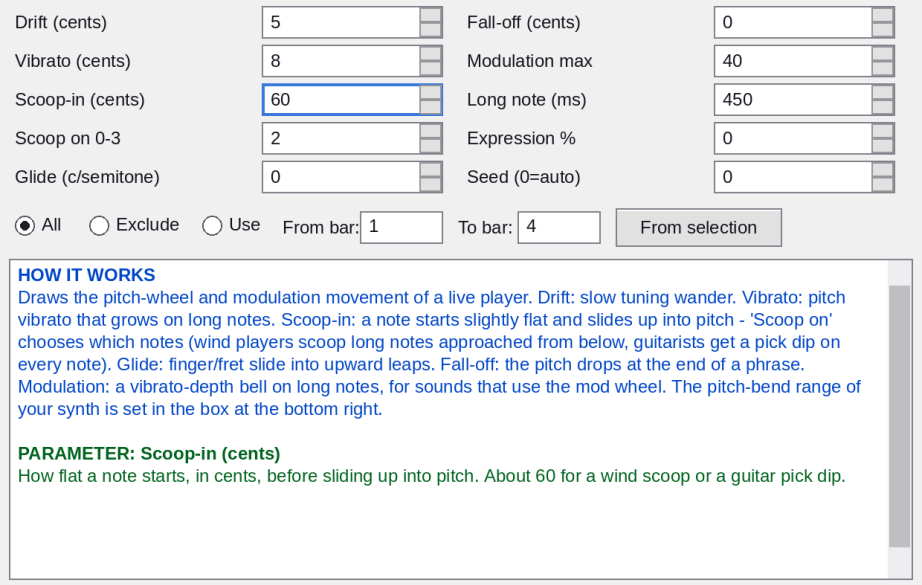


Figure - Human Pitch / Mod / Expr, default settings.

Parameter	Default	Range	What it does
Drift (cents)	5	0 to 100	Slow random tuning wander in cents (100 cents = 1 semitone). 3-6 is natural.
Vibrato (cents)	8	0 to 100	Depth of pitch vibrato on long notes, in cents. It fades in after a short delay. 0 = none.
Scoop-in (cents)	60	0 to 300	How flat a note starts, in cents, before sliding up into pitch. About 60 for a wind scoop or a guitar pick dip.
Scoop on 0-3	2	0 to 3	Which notes scoop: 0 = none, 1 = first note of each phrase, 2 = long notes approached from below (winds), 3 = every note (plucked strings).
Glide (c/semitone)	0	0 to 150	Extra slide into upward leaps of 3+ semitones, in cents per semitone of the leap (guitar fret glide about 20, violin 12). 0 = off.
Fall-off (cents)	0	0 to 300	Pitch drop on the last long note of a phrase, in cents (sax / trumpet 40-80).
Modulation max	40	0 to 127	Peak of the modulation-wheel (CC1) bell on long notes, 0-127. 0 = off.
Long note (ms)	450	50 to 5000	Only notes at least this long (milliseconds) get vibrato and modulation.
Expression %	0	0 to 100	Random expression (CC11) wobble in percent. 0 = leave CC11 alone (use Breath & Phrase Dynamics for expression).
Seed (0=auto)	0	0 to 999999	0 = automatic (stable for this effect). Any other number gives its own repeatable variation.

Draws the pitch-wheel and modulation movement of a live player. For the pitch to bend correctly, the **Bend range** box in the main window must match the pitch-bend range of your synthesizer (usually 2 semitones).

- **Winds:** Scoop on = 2 (long notes approached from below), Fall-off 40-80 cents for sax and trumpet.
- **Plucked strings:** Scoop on = 3 (every note, a pick dip) and some Glide (about 20 for guitar frets, 12 for violin).
- **Vibrato** grows on notes at least *Long note* milliseconds long and fades in after a short delay.
- Leave *Expression %* at 0 if you use Breath & Phrase Dynamics for the expression controller.

7.7 Breath & Phrase Dynamics

Shapes the loudness the way a wind player or bow does. Every note gets a breath attack that relaxes and swells toward the next note; every phrase rises and falls; and a breath is taken between phrases: the last note of the phrase is shortened (Breath gap) and the loudness dips during the gap. Velocity arc applies the same phrase shape to the note velocities. Output: 1 = Expression (CC11), 2 = Breath (CC2), 3 = both. Phrases come from the Phrase Finder above it in the chain.

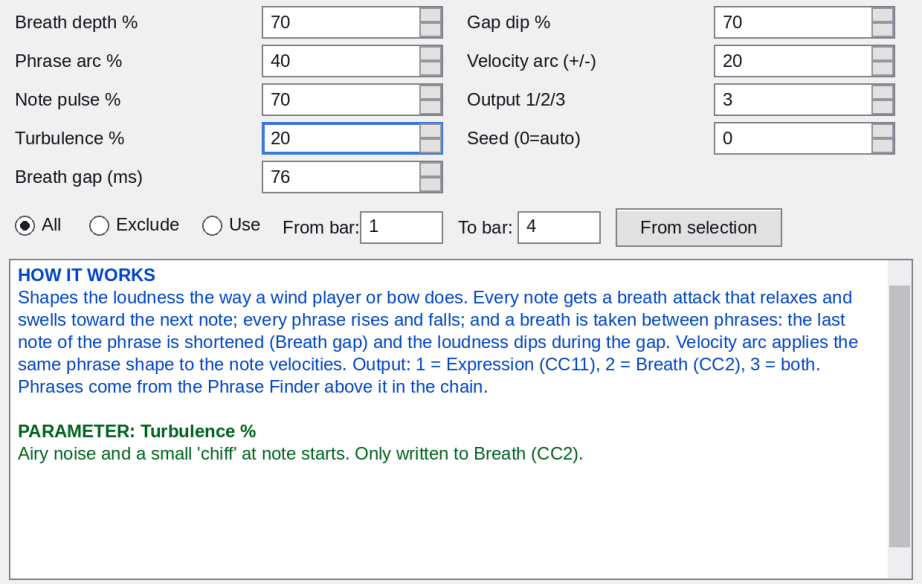


Figure - Breath & Phrase Dynamics, default settings.

Parameter	Default	Range	What it does
Breath depth %	70	0 to 100	How far the loudness moves, in percent of the controller range. 100 matches the reference flute (values from about 90 to 127).
Phrase arc %	40	0 to 100	How much the loudness rises and falls across each phrase.
Note pulse %	70	0 to 100	Strength of the attack / relax / swell shape inside every note.
Turbulence %	20	0 to 100	Airy noise and a small 'chiff' at note starts. Only written to Breath (CC2).
Breath gap (ms)	76	0 to 1000	The last note of every phrase is shortened so that this many milliseconds are left for the breath. 0 = keep note lengths.
Gap dip %	70	0 to 100	How far the loudness drops during the breath gap (70 = down to about a third, as in the reference flute).
Velocity arc (+/-)	20	0 to 127	How much the note velocities follow the phrase arc: softer at the start and end, strongest in the middle. 0 = keep velocities.
Output 1/2/3	3	1 to 3	1 = Expression CC11, 2 = Breath CC2, 3 = both.
Seed (0=auto)	0	0 to 999999	0 = automatic (stable for this effect). Any other number gives its own repeatable variation.

Shapes the loudness the way a wind player or a bow does: every note has an attack that relaxes and swells, every phrase rises and falls, and a breath is taken between phrases.

- Phrases come from the Phrase Finder above it in the chain. Without one, the default phrase settings are used.
- *Breath gap* shortens the last note of each phrase so that there is time for the breath; *Gap dip* lowers the loudness during that gap.
- *Output* chooses which controller is written: 1 = Expression (CC11), 2 = Breath (CC2), 3 = both. Use the controller your instrument sound responds to.

- *Velocity arc* applies the same phrase shape to the note velocities: softer at the start and the end, strongest in the middle.

7.8 Strum / Chord spread

Spreads the notes of a chord in time like a strummed or arpeggiated instrument (guitar, harp, accordion, piano). Notes that start within the chord window count as one chord and are delayed one after the other. Direction picks down-strokes (low to high), up-strokes (high to low) or alternating strokes; Velocity taper makes the later notes a little softer.

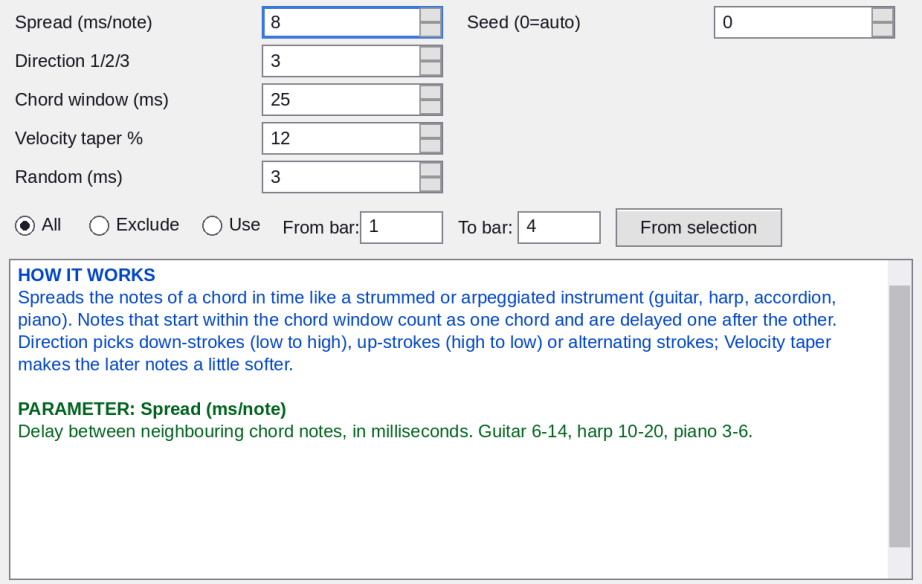


Figure - Strum / Chord spread, default settings.

Parameter	Default	Range	What it does
Spread (ms/note)	8	0 to 200	Delay between neighbouring chord notes, in milliseconds. Guitar 6-14, harp 10-20, piano 3-6.
Direction 1/2/3	3	1 to 3	1 = down-stroke (low to high), 2 = up-stroke (high to low), 3 = alternate.
Chord window (ms)	25	1 to 500	Notes starting within this many milliseconds of each other are treated as one chord.
Velocity taper %	12	0 to 100	How much softer the last note of the strum is than the first, in percent.
Random (ms)	3	0 to 200	Extra random delay per note in milliseconds, so the strum is not mechanical.
Seed (0=auto)	0	0 to 999999	0 = automatic (stable for this effect). Any other number gives its own repeatable variation.

Spreads the notes of a chord in time like a strummed or arpeggiated instrument.

- Typical spread (ms per note): guitar 6-14, harp 10-20, piano 3-6.
- Notes starting within the *Chord window* count as one chord. *Direction*: 1 = down-stroke (low to high), 2 = up-stroke, 3 = alternating.
- *Velocity taper* makes the later notes of a strum a little softer; *Random* adds a small random delay so the strum is not mechanical.

8. A complete example

The picture below shows one track of a real song, first as it was in the file and then after a chain of five effects: Phrase Finder, Velocity (95%), Humanize, Human Pitch / Mod / Expr and Breath & Phrase Dynamics. The top strip is quantized data with flat velocities. In the lower strip you can see the added modulation swells (violet), the pitch bends at the start of notes (orange), the expression curve that rises and falls with each phrase (light blue), and the slightly changed velocities (pink). The thin grey lines and outlines are the original, drawn as a ghost behind the result.

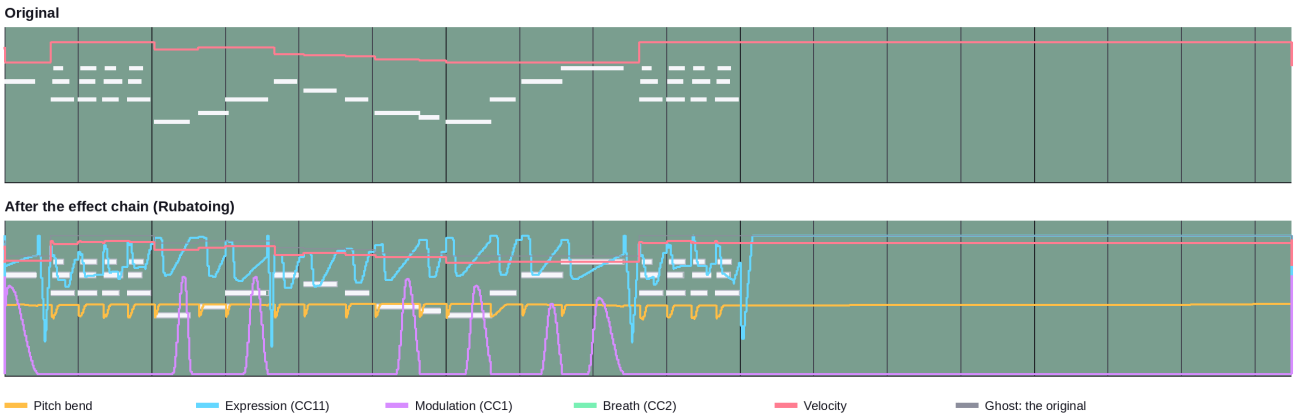


Figure 4 - The same track before and after the chain (illustration rendered from real data).

A good working method

1. Start with a factory preset close to your instrument. Listen to a few bars with **Loop** on.
2. Press **Bypass FX** to compare with the original, and look at the red **Rubatoing** / **Original** indicator to see which state you are hearing.
3. Click into a parameter box to read what it does, and change one parameter at a time. Playback restarts by itself, so you hear every change.
4. Keep the effect modest. A little goes a long way: typical *Humanize* timing values are only a few milliseconds.
5. When you are happy, **Apply All FXs**, then **Export MIDI file**. Save the project too, so you can come back to the chains.
6. Import the exported MIDI file into your DAW and assign your instrument sounds to it.

Saving, exporting and undoing

- **Save project (*.rpj)** keeps everything: song, chains, mixer, markers. It works even if the effects have not been applied.
- **Export MIDI file** writes exactly what you see and hear as a format 0 file, with the effects applied to the tracks that have chains, the mixer settings, and your markers.
- **Apply All FXs** makes the effects part of the song in the program. **Undo** goes back to the state before the last Apply, or brings back a deleted track.
- **Bypass FX** switches all effects off temporarily, for A/B listening. It does not delete anything.

9. Rubato for Korg Style creators

Arranger keyboards such as the Korg Pa series play **styles**: accompaniment patterns for drums, bass and chord instruments, with an intro, variations, fills, a break and an ending. Many style creators start from an existing MIDI file. The trouble is that quantized MIDI sounds exactly like what it is: every note on the grid, every velocity the same. Played through a good sound set, it still sounds like a machine.

Rubato can be used by Korg Style creators to convert existing MIDI into realistic, human-played backing tracks. You prepare the MIDI file in Rubato (choose the useful tracks, set the channels, mark the sections, and add the human touch), finish the file in your DAW, and import it into the Korg Style editor. The style that you hear on the keyboard then has the timing, dynamics and expression of a live player.

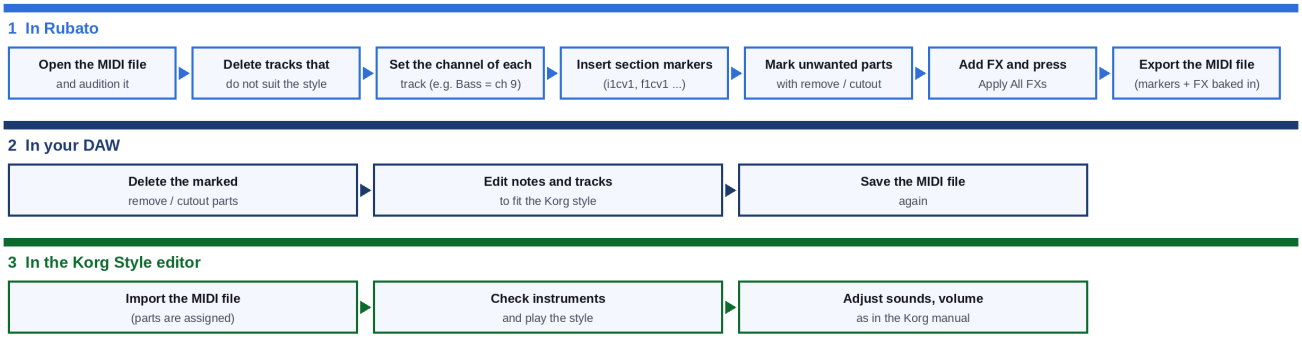


Figure 5 - The workflow at a glance: prepare in Rubato, finish in a DAW, build the style in the Korg Style editor.

> **What Rubato does not do.** Rubato deliberately does **not** edit or delete individual notes. That is not the purpose of the program. Everything that needs note-level editing (removing a wrong note, changing a chord, fixing a bar) is done in your DAW in step 7. In Rubato you work with whole tracks, whole sections and the human-feel effects.

Step 1 - Listen to the MIDI and delete tracks that do not suit the style

Open the MIDI file (Ctrl+O) and press Play. Use **Mute** and **Solo** on the track rows to listen to the parts one at a time, and decide what makes sense for the style: a style needs a drum pattern, a bass line and a few chord or accompaniment parts, not a full song arrangement with melody, solos and effects tracks.

To remove a track, right-click its row and confirm. Rubato asks first, and **Undo** (Ctrl+Z) brings the track back if you change your mind. Only whole tracks are deleted; the individual notes of the tracks you keep are left as they are.

Step 2 - Assign the channels that the Korg Style needs

A Korg style assigns every part of the accompaniment to a fixed MIDI channel. Use the **Ch** box on each track row to give every element the channel that the style expects. For example, if the bass is assigned to channel 9, find the bass part, click its Ch box and choose 9. Do the same for the drums, the chord parts and the other elements.

> **Check the channel table of your instrument.** The channel assignments differ between Korg models. Keep the channel table of the Style Creation chapter of the Korg manual at hand and work through it track by track. Rubato shows channels as 1 to 16, like the Korg display.

Step 3 - Insert the section markers

The Korg Style editor finds the sections of the style (intro, variations, fills, break and ending) from **markers** in the MIDI file. Rubato's Marker Track writes them directly into the MIDI file, at exactly the position you choose.

1. Click the **ruler** at the start of the section. The start position snaps to a beat, and the information panel shows the position as bar:beat:tick. Zoom in if you need to be more precise.
2. Choose *Edit* > *Insert marker at the start position*, or double-click the Marker Track, or right-click it and choose *Insert marker here*.
3. Type the marker text (letters and digits only) and press OK. A small orange flag with the text appears on the Marker Track.

For example, position the playhead at 1.1.0 and add the marker **`i1cv1`** for Intro 1; position the playhead at 6.1.0 and add the marker **`f1cv1`** for Fill 1; and so on for the variations, the break and the ending.

Section	Example marker	Example position
Intro 1	i1cv1	1.1.0 (bar 1, beat 1, tick 0)
Fill 1	f1cv1	6.1.0 (bar 6, beat 1, tick 0)
Variations, break, ending and the other intros and fills	Use the marker names listed in the Korg manual for your instrument, following the same pattern	The first beat of the section

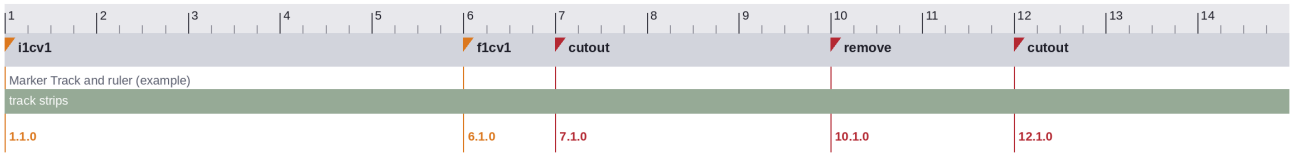


Figure 6 - Example of a Marker Track: Intro 1 at 1.1.0, Fill 1 at 6.1.0, and cut-out markers on unwanted parts (illustration).

> **Good to know.** There can be only one marker at a position; inserting at a position that already has a marker opens that marker for editing. Double-click a marker to rename it, right-click it to delete it, and use *Delete all markers* in the same menu to start over. The markers are saved in the project file as well, so you can come back to them later.

Step 4 - Mark the parts that are not useful

Most MIDI songs contain parts that the style does not need: a long verse, a solo, a different ending. Because Rubato does not delete notes, mark those parts so that they are quick to find and remove later in your DAW. Insert markers with the text **`remove`** or **`cutout`** at the start of each unwanted part. A suggestion: put the next useful marker (for example the next section marker) where the unwanted part ends, so that every stretch between two markers is clearly either useful or marked for removal.

Step 5 - Apply the human touch (FX)

If you have not yet added effects to the tracks, do it now as described in sections 5 and 7. Select a track, open its MFX Chain window (Ctrl+E) and load a preset that fits the instrument (plucked bass, guitar, piano, strings, and so on), or build your own chain. For drums and percussion, use light timing and velocity effects (Humanize, Velocity) only. Listen to the whole track in a loop, and compare with the original with **Bypass FX**.

> **Think of the loop.** A style is a short pattern that the keyboard plays over and over. Rubato bakes the human feel into the notes *once*, so every repeat of the pattern has exactly the same feel. Keep the effects moderate, and prefer a longer section (two or four bars) over a one-bar loop if you want it to sound more alive. Keep timing changes small, so that the first notes of a section stay close to its marker.

Do not forget the final step of the preparation: press **Apply All FXs** (toolbar, or *MIDI Process* > *Apply all FXs to the song*). This writes the effects into the MIDI data. The indicator at the bottom right reads *Original* again, because the effects are now part of the song.

Step 6 - Export the MIDI file

Choose *File > Export MIDI file* (Ctrl+Shift+S) and give the file a name. The exported file contains **all the markers** and has **the effects baked in**. Rubato exports a format 0 MIDI file (one track with all the channels); your DAW can show or split it by channel. Also save the project (Ctrl+S), so that you can return to the chains and markers if you want to try a different variation.

Step 7 - Finish the file in your DAW

Open the exported MIDI file in the DAW of your choice. The markers appear on its marker or arrangement track.

- Find the sections marked `remove` or `cutout` and delete them.
- Do any note-level editing that the style needs (this is where individual notes are corrected or deleted).
- Adjust the tracks so that they conform to the Korg style: parts on the right channels, the sections at the positions where the markers are.
- Save the MIDI file again from the DAW, in the MIDI format that the Korg manual asks for.

Step 8 - Import into the Korg Style editor

Import the new MIDI file into the Korg Style editor. All the parts are processed and assigned to the style. Check that the correct instruments are assigned to the parts, then play back the style. You should now have a style that plays with a human touch.

Step 9 - Finish the style on the instrument

Follow the Korg manual for your instrument to make the other changes: choose and adjust the sounds, set the playback volumes, and so on.

Checklist

Where	Done?	Task
Rubato	[ ]	Unwanted tracks deleted (right-click the track row)
Rubato	[ ]	Every track on the channel that the style needs (Ch box)
Rubato	[ ]	Section markers inserted (for example i1cv1 at 1.1.0, f1cv1 at 6.1.0)
Rubato	[ ]	remove / cutout markers on the unwanted parts
Rubato	[ ]	FX added to the tracks, and **Apply All FXs** pressed
Rubato	[ ]	MIDI file exported; project saved
DAW	[ ]	Marked parts removed, notes and tracks adjusted, MIDI saved again
Korg Style editor	[ ]	MIDI imported, instruments checked, style played back
Korg instrument	[ ]	Sounds and volumes set as in the Korg manual

Good luck with your humanized style creation!

10. Questions and answers

What exactly does Rubato do to my MIDI files?

It analyzes the rigid structure of quantized MIDI data and applies models that mimic human muscle memory, physical instrument constraints and musical phrasing. It shifts note start times, durations and velocities and draws pitch-bend, modulation, breath and expression curves to recreate the subtle imperfections of a live performance.

Is Rubato a standalone program or a VST plug-in?

Rubato is a standalone program for Windows. It is very frugal with CPU resources and runs on machines with low specifications. Open or drag in MIDI files, process them, and export the result to your DAW. To hear the song while you work, choose a synthesizer or virtual MIDI port in the *MIDI out* box.

Does it work with my DAW?

Yes. Rubato exports standard MIDI data, so it works with any DAW that handles MIDI, including FL Studio, Ableton Live, Cubase, Studio One, Reaper and Pro Tools.

How do the instrument presets work?

Different instruments have different physical limitations: a guitarist cannot jump five octaves instantly, and a saxophone player varies wind pressure across a single breath. The presets combine the effects with values tuned for specific instruments, so a guitar track behaves like strings and frets and a horn track behaves like lungs and brass.

Will this ruin my song’s timing or sync?

No. Individual notes flex, rush or drag, but the tempo map is not changed, so the phrases stay anchored to the project tempo and tracks never drift out of time.

I do not hear any difference.

Check that the indicator at the bottom right says **Rubatoing** (a track has an effect) and that *Bypass FX* is off. Then check that the effect is not muted in the Mute column, and that the parameters are not zero. For pitch effects, make sure that your instrument sound responds to pitch bend and modulation. For breath effects, make sure it responds to the controller chosen in *Output* (CC11 or CC2).

The pitch bends sound too large or too small.

The *Bend range* box in the main window must match the pitch-bend range of your synthesizer. Most instruments use 2 semitones; some use 12 or 24.

I opened a file and it has only one track.

A format 0 file has all channels in one track. Rubato splits it into one track per channel when it opens it, so each instrument has its own strip and chain.

What are the Windows system requirements?

Windows 10 or Windows 11 (64-bit). It needs very little CPU and RAM. It may run on earlier versions of Windows, but that has not been tested.

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